

MSc Robotics, Boston University · graduating December 2026, available January 2027

German citizen · three years of US work eligibility via OPT and the STEM extension

EXPERIENCE

2026 – present	Graduate Researcher, Robotics MBZUAI · Abu Dhabi, operated from Boston - Built a full-stack teleoperation system operating Boston to Abu Dhabi over the public internet — self-developed low-latency H.264/RTP video with FEC and adaptive quality control, plus per-device UDP state streams and a reliable acknowledged channel for commands that must not be dropped. - Bimanual VR teleoperation: Cartesian impedance control with nullspace posture regulation, resolved-rate IK with damped least squares, pan-tilt joint impedance, and a control-barrier-function self-collision filter — running interchangeably against MuJoCo and libfranka. Unreal Engine 5 VR interface driven entirely by gaze and speech with an attention-adaptive layout, keeping both hands on the arms, plus a live HUD of measured end-to-end latency, jitter and loss. - Collected and labelled a bimanual parcel-sorting dataset for intent recognition and imitation learning (ACT). - Research question: does guiding an operator during collection produce demonstrations that train better policies? Paired policy comparison in progress.
04/2024 – 2025	Researcher — Teleoperation and Shared Control MIRMI, TU Munich · Haddadin Lab - Built a distributed real-time robotic control stack over a local network: 7-DoF robot arm, pan-tilt head, and passivity-based haptic feedback control. - Ran a ten-participant, six-condition, ~300-trial user study on operator guidance — assistance cut NASA-TLX workload by over 40 points and median contact force from ~21 N to ~12 N with no cost in completion time. - Co-authored "Enhanced Robotic Navigation in Deformable Environments using Learning from Demonstration and Dynamic Modulation," IEEE/RSJ IROS 2025.
08/2022 – 04/2023	Intern, Motor Controls Tesla · System Architecture, Modeling & Controls Automated regression test tooling validating control algorithms across multiple thermal motor models; Python post-processing for test data analysis.
2018 – 2022	Technical Director, then Head of Drivetrain Development Formula Student (Formula SAE) · Munich University of Applied Sciences Technical director for ~60 people across seven subteams (two years): vehicle concept, cross-subsystem integration, budget, and sponsors including BMW, Knorr-Bremse and Infineon. - Designed a 20,000 rpm, 29 Nm traction motor, wrote its field-oriented control, and built the test bench that characterised it.

EDUCATION

2026	MSc Robotics — Boston University. <i>Expected December 2026.</i>
2023 – 25	MSc Controls & Automation — Technical University of Munich. Thesis at MIRMI, Haddadin Lab on operator nudging and shared control in teleoperation.
2018 – 22	BSc Electrical Engineering , electromobility focus — Munich University of Applied Sciences.

SELECTED PROJECTS

POLICY	Force-domain diffusion policy for sub-millimetre insertion. Contact-rich insertion from force alone, no vision. DDIM-10 reached 78% against the scripted expert's 61%, and transferred zero-shot to unseen peg geometries.
CONTROL	Vision-based NMPC for pushing under perception uncertainty. ArUco/PnP at 60 Hz feeding an EKF and acados SQP-RTI NMPC at 100 Hz; nominal, certainty-equivalent and chance-constrained control compared over a 360-run sweep.

SKILLS

CONTROL	Nullspace redundancy resolution, resolved-rate IK, control barrier functions, NMPC, LQR / iLQR, EKF
LEARNING	Imitation learning (ACT), diffusion policies (DDPM / DDIM), causal sequence models (GRU, transformer), reinforcement learning
SOFTWARE	C++17 (real-time), Python, MATLAB / Simulink · MuJoCo, libfranka, Pinocchio, acados, CasADi, PyTorch, Unreal Engine 5, OpenXR, GStreamer, CMake, Linux, Git · German (native), English (fluent)